A four-phase, positive-sheet reconstruction

Complex Hadamard matrices of order six

Scope. The map below covers, up to row/column permutations and unit-phase scalings, **every order-six Hadamard class except Tao and F_6**. Its domain is precisely $\{(a,b,c,d) \in \mathbb{T}^4 : \mathcal{R} > 0\}$, where $\mathbb{T} = \{z : |z| = 1\}$. It is a **surjective covering, not a one-to-one parametrization**: different corners can represent the same class. Matrices are unnormalized: $|H_{ij}| = 1$ and $HH^\dagger = 6I_6$.

1. Seeds and shared radical. Put

$$E = \begin{pmatrix} 1 & 1 & 1 \\ 1 & a & b \\ 1 & c & d \end{pmatrix}, \qquad G_h = 6I_3 - EE^\dagger.$$

The horizontal and vertical data are

$$\begin{aligned} s_h &= -(1+a+b), & t_h &= -(1+c+d), \\ r_h &= -(1+c/a+d/b), & & \\ s_v &= -(1+a+c), & t_v &= -(1+b+d), \\ r_v &= -(1+b/a+d/c). & & \end{aligned}$$

Using $(s,t,r) = (s_h, t_h, r_h)$, define

$$\begin{aligned} X = |s|^2, \quad Y = |t|^2, \quad Z = |r|^2, \\ p = X+Y+Z, \quad J = 2\Re(sr\bar{t}), \\ W = XY + XZ + YZ - XYZ, \\ g = 27 - 3p + J, \quad \eta = 9 - p + J, \\ \chi = W - (p-9)(p-18) + (p-15)J, \\ \mathcal{R} = 4g^2\eta - \chi^2 > 0, \\ z_h = \frac{\chi + i\sqrt{\mathcal{R}}}{2g}, \qquad z_v = \bar{z}_h. \end{aligned}$$

The same $g, \eta, \chi, \mathcal{R}$ result from the vertical data. Take $\sqrt{\mathcal{R}} > 0$.

2. Side recipe. Evaluate the following twice: at (s_h, t_h, r_h, z_h) and at (s_v, t_v, r_v, z_v).

$$\begin{aligned} P_0 &= rs^2 + \bar{r}t^2 + (3 - |r|^2)st, \\ P_1 &= t\bar{s}^2 + r^2\bar{t} + (3 - |t|^2)r\bar{s}, \\ P_2 &= s\bar{t}^2 + \bar{r}^2\bar{s} + (3 - |s|^2)\bar{r}\bar{t}, \\ (c_0, c_1, c_2) &= (9 - |r|^2, 9 - |t|^2, 9 - |s|^2), \\ \lambda_j &= \frac{P_j(g + c_j\bar{z})}{|P_j|^2} \quad (j = 0, 1, 2). \end{aligned}$$

Continue the same recipe with

$$\begin{aligned} u = \lambda_0\bar{\lambda}_1, \quad \mu = s/3, \quad \nu = t/3, \\ n = 1 - |\mu|^2, \qquad k = r/3 - \nu\bar{\mu}, \\ \alpha = \mu^2 - u\bar{\mu}, \quad \gamma = 2\mu\nu - z\lambda_0/3, \\ \delta = \sqrt{\frac{g}{27}\lambda_0\bar{\lambda}_1\bar{\lambda}_2}, \\ d_F = \frac{27}{g}(n\gamma - 2\alpha k)\bar{\delta}. \end{aligned}$$

The number d_F is real. Choose the sign of δ so that $d_F \geq 0$; either sign is allowed at equality. Set

$$\begin{aligned} \rho = (n + d_F)/2, \qquad q = \alpha/\rho, \\ c_+ = (k - \delta\bar{q})/n, \quad c_- = (kq + \delta)/n, \\ \sigma = u + \mu q, \qquad \xi = \sigma^2/\rho, \\ M = \begin{pmatrix} 1 & 0 & 0 \\ \mu & \sigma & q\bar{\sigma} \\ \nu & c_+\sigma & c_-\bar{\sigma} \end{pmatrix}, \qquad w = \sqrt[3]{\xi}. \end{aligned}$$

This gives (M_h, w_h) and (M_v, w_v). Here $|\xi| = 1$; any cube root may be used. Its three choices only cycle the corresponding last three rows or columns. The square-root tie is also a side-permutation choice.

3. Assemble $H_+(a,b,c,d)$. Define

$$\omega = e^{2\pi i/3}, \quad F_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix},$$

$$\begin{aligned} P(w) &= \operatorname{diag}(1, \bar{w}, w), \\ B &= M_h P(w_h) F_3, \\ C &= F_3 P(w_v) M_v^T, \\ D &= -\frac{1}{g} CE^\dagger \operatorname{adj}(G_h) B. \end{aligned}$$

$$\boxed{H_+(a,b,c,d) = \begin{pmatrix} E & B \\ C & D \end{pmatrix}.}$$

The adjugate is the transpose of the cofactor matrix: its entries are signed 2×2 determinants. Thus this is an explicit radical formula entirely in the four seeds. The symbol $\dagger$ denotes conjugate transpose; T denotes transpose.

Validity and coverage. On $\mathcal{R} > 0$, $g > 0$, all $P_j \neq 0$ and $n, \rho > 0$; no additional guards or flatness test are needed for this matched pairing. A negative-sheet representative can be moved to its opposite corner and dephased to reverse the sheet while preserving $\mathcal{R} > 0$. Hence one positive pairing suffices globally, although it does not enumerate every completion of a prescribed corner. There are three square roots and two phase cube roots per reconstruction, with no further continuous parameters.

External proof inputs. Strict-corner coverage, matched-completion exhaustiveness (including the exceptional Tao pairing), and opposite-corner sheet reversal are taken as proved elsewhere by the collaborator; they are not proved or independently certified in this note. Sources: *Comprehensive Handoff*, §9 (21 September 2026); *Single-Sheet Conjugate Fourier Form*, §6; and *Unitary Core Phase Gate Form*.